Week 1

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)
- Explain the role of probability in statistics. (CLO 4)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe statistics as the process of collecting, describing, and analyzing data. (CLO 1) **2.** Determine whether a surprising result could reasonably have happened by chance alone. (CLO 4) **3.** Distinguish between a sample and a population. (CLO 1) **4.** Explain how a non-representative sample can bias an estimate. (CLO 1) **5.** Compute and compare the mean and median of a small dataset, and explain how an outlier affects each. (CLO 2)	START HERE: Welcome; Syllabus; Textbook and Wiley Course Resources; Online Tutoring; College Policies & Student Support Services Introduction Slides: What Is Statistics For? (Day 1)	Participation: Syllabus – course tour and expectations Day-1 class data activity: collect everyone's age, then find the mean, median, mode and boxplot together; compare the class median to the U.S. median age	Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	Blackboard Zoom Respondus
6. Identify the cases and variables in a dataset presented as a table. (CLO 2) **7.** Classify a variable as categorical or quantitative. (CLO 2) **8.** Identify the explanatory and response variables in a research question. (CLO 1)	e-Text: 1.1 The Structure of Data Interactive Lecture: 1.1 The Structure of Data Slides: 1.1 The Structure of Data	Participation: Variables – identify cases, variables and variable types Your Turn: Used Cars: Cases and Variables	Online Homework: 1.1 The Structure of Data (due Sun Aug 30, 11:59 PM) Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus